Q1: Discuss India's plan to establish its own space station and identify the key advancements required for the successful development and operation of this space station. (10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about Space stations and current status of functional space stations.
Body: Address the demand by highlighting India's plan about its own space station and about the required advancement.
Conclusion: Conclude with summarizing the topic or a way forward.

Answer: A space station is a spacecraft designed to support crew members for extended periods and to facilitate docking with another spacecraft. Currently, there are two fully functional space stations in Earth's low orbit:

- **International Space Station (ISS)**: A collaborative effort among NASA (USA), JAXA (Japan), CSA (Canada), and Roscosmos (Russia), the ISS represents the world's largest international cooperative program in science and technology.
- **Tiangong Space Station**: China's space station, which is being developed as part of its space program.

India also aims to establish its own space station, with the Indian Space Research Organisation (ISRO) targeting the creation of the "**Bhartiya Antariksha Station" by 2035**. The proposed station is expected to orbit Earth at approximately 400 km in low Earth orbit (LEO) and accommodate astronauts for 15-20 days.

Required Advancements:

- **Gaganyaan Requirements**: India needs to meet all requirements for the Gaganyaan mission, including space suits, astronaut training facilities, a crew escape module, and the development of a habitable module.
- **Rocket Capability**: Upgrading the GSLV-MK-III, which currently carries up to 10 tonnes to LEO, to a larger and more capable rocket is crucial.
- **Space Docking Technology**: Developing the ability to perform space docking is vital. ISRO plans to conduct a space docking experiment, Spadex, involving two experimental modules launched aboard a PSLV rocket. Successful docking is essential for assembling larger structures in space and for future human missions.

Although the space station program is still in its early stages, ISRO has made significant strides by initiating human space missions and preparing for the Spadex experiment. These steps mark progress toward the ambitious goal of establishing India's own space station, positioning the country as a notable player in space exploration.

Additional Information:

Advantages of space stations:

- **Natural Progression**: Establishing a space station is a logical step for any space-faring nation following successful human spaceflight. It will enhance India's capabilities in human-rated rockets, astronaut training, life support, and crew safety.
- **Future Manned Expeditions**: The space station will pave the way for future manned missions to other celestial bodies.
- **Microgravity Experiments**: It will enable the conduct of experiments in microgravity, which are essential for various scientific and technological advancements.

Q2: Discuss the diverse applications of nanotechnology across various fields. Highlight the potential issues that need to be addressed for effective use of nanotechnology. (10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about nanotechnology.
Body: Address the demand by highlighting diverse application of nanotechnology followed by potential issues that need to be addressed.
Conclusion: Conclude with summarizing the topic or a way forward.

Answer: Nanotechnology is the science of manipulating matter at the atomic and molecular scale, typically in the range of 1-100 nanometres, has emerged as a transformative force across diverse sectors. It has revolutionized various fields by enabling innovations and advancements previously deemed impossible.

Diverse applications of Nanotechnology:
- **Healthcare:** Targeted drug delivery, early disease diagnosis, and tissue engineering with nanomaterials. **Agriculture:** Improved crop yields, food safety, efficient fertilizers, pest protection, and better soil quality.
- **Energy:** Enhanced solar cells, advanced batteries, improved fuel cells, and better energy storage.
- **Environment:** Water purification, soil remediation, air pollution control, and efficient renewable energy.
- **Electronics:** Miniaturization of devices, faster computers, flexible electronics, and wearable tech.
- **Space:** Create stronger spacecraft components and miniaturized sensors for exploration.
- **Transportation:** Develops lighter, fuel-efficient vehicles and advanced materials for automotive and aerospace use.

Potential issues that need to be addressed:
- **Environmental Impact:** Nanomaterials may be toxic and affect ecosystems, rigorous assessment is needed.
- **Health Risks:** Potential health effects from nanomaterials are under study, safety protocols are essential.
- **Ethical Concerns:** Nanotechnology raises issues like biological weapons and human enhancement, ethical frameworks are necessary.
- **Economic Inequality:** High costs may worsen economic disparities; efforts should ensure equitable access.
- **Regulatory Challenges:** Regulations lag behind rapid development, adaptable rules are required.
- **Security Risks:** Misuse for nano-weapons or surveillance poses threats, security measures are needed.

Nanotechnology holds immense promise for addressing global challenges. However, responsible development and utilization are crucial. A multidisciplinary approach is essential to harness its benefits while mitigating risks. By addressing potential issues proactively, we can ensure that nanotechnology contributes positively to society.

Q3: Discuss how Digital Public Infrastructure (DPI) can contribute to India's growth trajectory, and also address the concerns associated with it. Provide suitable illustrations to support your answer.
(10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about Digital Public Infrastructure (DPI).
Body: Address the demand by discussing how Digital Public Infrastructure (DPI) contribute to growth followed by associated concern.
Conclusion: Conclude with summarizing the topic or a way forward.

Answer: Digital Public Infrastructure (DPI) provides the technological backbone for enabling digital interactions and services which is crucial for India's inclusive and sustainable growth.

DPI's Contribution to Growth:
- **Countering Monopolistic Practices:** UPI mitigates data colonization by VISA, Google, enabling fair competition in digital payments.

- **Inclusive E-commerce:** Open Network for Digital Commerce (ONDC) facilitates digital onboarding of small retailers, promoting inclusive growth in the e-commerce sector.
- **Financial Inclusion:** JAM trinity, Aadhaar-based Direct Benefit Transfer (DBT) for subsidies enhance targeted delivery of benefits and democratize credit access, boosting financial inclusion.
- **E-governance:** CPGRAMS, Digital India Land Records Modernization Programme (DILRMP) ensures transparent and efficient public service delivery.
- **Digital Development:** Platforms like SWAYAM and Diksha provide digital education, while Ayushman Bharat Digital Mission integrates digital health infrastructure.
- **Modernizing Agriculture:** Agri-stack leverages digital tools for agriculture, and Digital India Land Records Modernization Programme enhances land record management.

Associated Concerns:

- **Digital Divide:** Marginalization of 'digital have nots' due to lack of devices for accessing digital services, such as education.
- **Infrastructure Gaps:** Urban bias in digital access due to inadequate electricity and internet connectivity in rural areas.
- **Privacy and Data Security:** DPI poses risks to individual privacy and data security.
  - For example: Concerns over Aadhaar data breaches.
- **Cybersecurity Threats:** Increased cyber-attacks can disrupt services.
  - For example: Recent ransomware attacks on critical infrastructure.
- **Surveillance Concerns:** Issues of privacy and state surveillance, exemplified by the Pegasus spyware controversy.

Measures to Address Concerns:

- **Bridging Digital Divide:** Campaigns for digital literacy and expanding rural broadband infrastructure. Use Community Service Centers for last-mile connectivity.
- **Data Protection Policies:** Implement personal and non-personal data protection laws with transparency, consent mechanisms, and accountability.
- **Cybersecurity Enhancements:** Continuous monitoring, risk assessment, and public-private partnerships to bolster cybersecurity resilience.
- **Competitive Digital Ecosystem:** Promote open standards, interoperability, and fair market access to foster competition.

Balancing digital empowerment with safeguarding rights, privacy, and security ensures DPI catalyzes inclusive growth, innovation, and sustainable development in India.

Q4: Highlight the impact of marine pollution on the coastal biodiversity of India. What steps can be taken to mitigate this pollution and protect this biodiversity? (10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about India's coastal ecosystem.
Body: Address the demand by discussing Impact of marine pollution on the coastal biodiversity followed by Steps taken to mitigate marine pollution.
Conclusion: Conclude with summarizing the topic or a way forward.

Answer: India features a diverse array of coastal ecosystems, including coral reefs, mangroves, lagoons, and estuaries. However, this biodiversity faces a significant threat from marine pollution.

Impact of marine pollution on the coastal biodiversity

- Habitat destruction through pollutants degrades essential ecosystems, disrupting marine life and causing species loss.
- Coral Reef Degradation due to runoff nutrients and sediments into the oceans.

- According to UNEP, approximately 25% of the world's coral reefs have been destroyed, and another two-thirds are under severe threat from pollution, overfishing, and climate change.

- Bioaccumulation as heavy metals and persistent organic pollutants accumulate in marine organisms, causing threat to their life.
- Ocean acidification due to increased CO_2 emissions which harms shell-building animals and other marine life.
- Eutrophication results from excessive nutrient runoff from agriculture and wastewater discharge.

Steps taken to mitigate marine pollution

- Understanding the source, distribution, and dynamics of waste is important to target priority areas.
- Sustainable agricultural practices that minimize the use of fertilizers.
- Improved Wastewater treatment facilities to remove nutrients from industrial and domestic sewage before it is discharged.
- Regular beach cleanup campaigns such as 'Swachh Sagar Abhiyaan'.

Marine pollution poses a grave threat to coastal biodiversity. By implementing above mentioned steps, India can work towards protecting the valuable coastal ecosystems. This will also aid India in completion of Sustainable Development Goals 14 (Life below water).

Q5: What are the natural and anthropogenic factors that trigger landslides? Suggest measures to mitigate the issue of landslides. (10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about recent landslide incident in India
Body: Address the demand by highlighting natural and anthropogenic factors triggering landslides followed by measures to mitigate them.
Conclusion: Conclude with summarizing the topic.

Answer: Recently, the tragic Wayanad landslide claimed over 250 lives, emphasizes how climate change and unsustainable development practices can intensify landslide occurrences in vulnerable regions.

Natural and Anthropogenic Factors Triggering Landslides
Natural Factors:

- **Geological Conditions:** Weak or weathered rock layers, fault lines, and steep slopes are naturally prone to landslides.
- **Water Saturation:** Heavy rainfall, or the presence of water bodies can saturate the soil, increasing its weight and reducing its stability.
- **Seismic Activity:** Earthquakes and tremors can trigger landslides by shaking and de-stabilizing the ground.
- **Erosion:** Natural erosion by rivers, glaciers, or coastal processes can destabilize slopes.

Anthropogenic (Human-Induced) Factors:

- **Deforestation:** The removal of vegetation, which stabilizes the soil with its root systems, can lead to increased soil erosion and landslides.
- **Construction and Urbanization:** The construction of roads, buildings, and other infrastructure can alter the natural slope stability, increasing the risk of landslides.
- **Mining and Quarrying:** These activities can destabilize slopes by removing large amounts of earth and rock.
- **Water Management:** Poor water management practices, such as improper drainage systems or dam construction, can alter natural water flows and lead to landslides.

Measures to Mitigate Landslides

- **Afforestation and Reforestation:** Planting trees and vegetation can help stabilize the soil and prevent erosion.
- **Slope Stabilization Techniques:** Use of retaining walls, rock bolting, and terracing can help stabilize slopes and prevent landslides.
- **Proper Land Use Planning:** Avoiding construction in landslide-prone areas and implementing strict zoning laws can reduce the risk.
- **Improved Water Management:** Proper drainage systems, controlled irrigation, and management of water bodies can prevent water accumulation and soil saturation.
- **Early Warning Systems:** Implementing monitoring systems and early warning mechanisms can help predict and respond to landslide threats.
- **Geotechnical Surveys and Assessments:** Conducting detailed geological and geotechnical surveys before construction projects can identify potential risks and guide mitigation strategies.

By addressing both natural and anthropogenic factors, a comprehensive approach can be developed to reduce the risk and impact of landslides.

Q6: The devastating urban floods in India have highlighted the complex interplay between climate change, urbanization, and infrastructural deficiencies. Discuss (10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about recent urban flood incident in India
Body: Address the demand by highlighting complex interplay between climate change, urbanization, and infrastructural deficiencies followed by mitigation strategies.
Conclusion: Conclude with summarizing the topic.

Answer: The devastating urban floods in Indian cities like Mumbai, Chennai, and Bengaluru have brought to the forefront the intricate and interconnected factors of climate change, rapid urbanization, and infrastructural deficiencies.

Complex interplay between climate change, urbanization, and infrastructural deficiencies

- **Climate Change:** Recent studies indicate that climate change is significantly altering rainfall patterns in India, leading to increased erratic and extreme weather events. This change has manifested in higher occurrences of heavy rainfall, which is a primary contributor to urban flooding in cities like Bengaluru and Mumbai
- **Urbanization:**
    - **Rapid Expansion and Unplanned Development:** The rapid growth of urban areas, often unplanned, has led to the encroachment of natural drainage channels, wetlands, and floodplains. This reduces the land's natural ability to absorb excess rainwater.
    - **Increased Impervious Surfaces:** The use of concrete and asphalt in urban areas reduces the permeability of the ground, preventing rainwater from seeping into the soil and increasing surface runoff.
- **Infrastructural Deficiencies:**
    - **Insufficient Water Management:** Poorly managed water resources, including ineffective sewage systems and inadequate rainwater harvesting, contribute to the challenge of managing urban floods.
    - **Poor Urban Planning:** The lack of integrated urban planning and zoning regulations often results in construction in vulnerable areas, such as low-lying regions and reclaimed land, increasing the risk of flooding.

Mitigation and Adaptation Strategies:

- **Enhancing Urban Planning and Zoning:** Strengthening urban planning processes to prevent construction in flood-prone areas and promote the development of green spaces that can absorb rainwater.
- **Upgrading Infrastructure:** Investing in modern, efficient drainage and sewage systems, along with regular maintenance, to ensure they can handle heavy rainfall events.
- **Implementing Sustainable Practices:** Encouraging the adoption of sustainable urban development practices, such as rainwater harvesting, green roofs, and permeable pavements, to reduce surface runoff.
- **Climate Resilience Planning:** Integrating climate resilience into city planning, including early warning systems and disaster preparedness plans, to better anticipate and respond to flooding events.

Effective solutions will need a combination of robust planning, investment in infrastructure, and sustainable urban development practices to build resilient cities capable of withstanding future climate impacts.

Q7: Analyse how urban heat island (UHI) exacerbate the effects of climate change in urban areas. Suggest measures to mitigate the effects of UHI. (10 Marks, 150 Words)

Approach to the answer:
Introduction: Start with introducing in brief about Urban Heat Island effect.
Body: Address the demand by highlighting how UHIs exacerbate the effects of climate change in various ways followed by measures to mitigate effects of UHI.
Conclusion: Conclude with summarizing the topic.

Answer: The Urban Heat Island (UHI) effect refers to the phenomenon where urban areas experience higher temperatures than their rural surroundings, primarily due to human activities and urban infrastructure.

UHIs exacerbate the effects of climate change in several ways:

- **Increased Energy Demand:** Higher energy demand, often met by fossil fuel-based power plants, contributing to higher greenhouse gas (GHG) emissions. The increased GHG emissions further accelerate climate change.
- **Amplified Heat Waves:** UHIs can intensify the severity and duration of heat waves, making them more dangerous. Elevated nighttime temperatures reduce the cooling-off period, posing health risks.
- **Air Quality Degradation:** Higher temperatures can enhance the formation of ground-level ozone and other pollutants, exacerbating air quality issues.
- **Impact on Water Resources:** Increased temperatures can lead to higher rates of water evaporation from reservoirs, rivers, and lakes, exacerbating water scarcity issues.

Measures to Mitigate the Effects of UHI

- **Increasing Green Spaces:** Expanding urban green spaces, such as parks, green roofs, and community gardens, can help absorb heat, provide shade, and reduce the ambient temperature.
- **Cool Roofs and Pavements:** Implementing cool roofs and cool pavements can reflect more sunlight and absorb less heat. This reduces the heat island effect and lowers cooling energy demand.
- **Urban Planning:** Urban planning policies can promote the development of compact, mixed-use neighborhoods with ample green infrastructure.
- **Energy Efficiency Improvements:** Promoting energy efficiency in buildings, including better insulation and energy-efficient appliances, can reduce the energy demand for cooling. Transitioning to renewable energy sources can also decrease GHG emissions.
- **Water Management:** Implementing water-efficient landscaping and rainwater harvesting can reduce water evaporation and manage urban water resources more effectively.
- **Heat Action Plans:** Developing and implementing heat action plans can help cities prepare for and respond to heat waves. These plans may include early warning systems, public cooling centers, and emergency response strategies.

- **Adopting Smart Technologies:** Utilizing smart technologies for urban management, such as smart grids and sensor networks, can optimize energy use and monitor environmental conditions, helping to manage the UHI effect.

By implementing these measures, cities can mitigate the UHI effect, enhance urban resilience, and reduce the exacerbation of climate change impacts. A comprehensive approach combining policy, technology, and community engagement is essential to effectively address the challenges posed by UHIs.

Q8: Discuss the causes of increasing frequency of extreme weather events in India. What steps can be taken to improve disaster preparedness and response to these events? (10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about reasons of increasing extreme weather events in India.
Body: Address the demand by discussing causes of increasing frequency of extreme weather events followed by steps taken to improve disaster preparedness and response to these events.
Conclusion: Conclude with summarizing the topic.

Answer: The increasing frequency of extreme weather events in India is due to a combination of climate change, urbanization, and deforestation.

Causes of increasing Frequency of Extreme Weather Events:

- **Climate Change**: Global warming is intensifying monsoons, heatwaves, and cyclones. Example: 2024 heat waves in Northern India, The 2020 Amphan cyclone in the Bay of Bengal, etc.
- **Urbanization**: Rapid urban expansion leads to poor drainage, increasing flood risk. Example: Chennai floods in 2015.
- **Deforestation**: Loss of forest cover disrupts rainfall patterns and increases landslide risks. Example: Uttarakhand landslides in 2022.
- **Unsustainable Agriculture**: Overuse of groundwater and chemical fertilizers disrupts soil health and water cycles.
- **Global Phenomena**: Events like El Niño and La Niña influence India's weather patterns, causing abnormal rainfall and temperatures.

Steps to Improve Disaster Preparedness and Response:

- **Enhanced Early Warning Systems**: Develop and deploy advanced technology for timely alerts about extreme weather events.
  - *Example*: The Indian Meteorological Department's Doppler radars provide early warnings for cyclones and heavy rainfall.
- **Community Training and Education**: Conduct regular drills and awareness programs to educate people on how to respond to disasters.
- **Strengthen Infrastructure**: Build resilient infrastructure to withstand extreme weather.
  - *Example*: Coastal areas in Odisha have improved cyclone shelters and reinforced coastal defenses.
- **Effective Land Use Planning**: Implement zoning regulations to avoid construction in high-risk areas like floodplains and landslide-prone zones.
- **Invest in Research and Technology**: Support research on disaster risks and invest in technologies for better prediction and response capabilities.
- **Coordination Among Agencies**: Ensure seamless coordination between government, NGOs, and local communities during disaster response and recovery.

Addressing extreme weather events in India requires coordinated efforts in climate action, urban planning, and community engagement to mitigate impacts and enhance resilience.

Additional Information:

Steps Taken to Improve Disaster Preparedness and Response:

- **United Nations Office for Disaster Risk Reduction:** Leads coordination for disaster risk reduction within the UN system.
- **Coalition for Disaster Resilient Infrastructure (CDRI):**
  - Promotes infrastructure resilience to climate and disaster risks.
  - Provides technical support and capacity-building.
  - Engages in collaborative research and knowledge management.
- **Sendai Framework for Disaster Risk Reduction 2015-2030:**
  - Focuses on preventing new risks, reducing existing risks, and increasing resilience.
  - Addresses disaster risk dimensions: exposure, vulnerability, and hazard characteristics.
- **Humanitarian Assistance and Disaster Relief (HADR):**
  - India acts as a first responder in regional and global HADR operations (e.g., Operation Rahat, Operation Dost).
  - Conducts HADR exercises to enhance stakeholder synergy and interoperability (e.g., Exercise Samanvay, Tiger Triumph).
- **Infrastructure for Resilient Island States (IRIS):** Promotes disaster and climate resilience for infrastructure in SIDS, launched at COP26.
- **Disaster Management Act 2005:**
  - Establishes National, State, and District Disaster Management Authorities.
  - Constitutes National Disaster Response Force (NDRF) for specialized disaster response.
  - Establishes National Institute of Disaster Management for capacity development.
  - Mandates disaster mitigation and response funds at various levels.
  - Develops comprehensive national policies and plans based on Sendai Framework and UN SDGs.

Q9: Organized crime is often both a product of and contributor to conflict in vulnerable states. Discuss
(10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about Organized crime.
Body: Address the demand by highlighting the factors of thriving organized crime in vulnerable states followed by what organized crime contributes.
Conclusion: Conclude with summarizing the topic or a way forward.

Answer: Interpol defines organized crime as "Any group having a corporate structure whose primary objective is to obtain money through illegal activities." Such activities include drug abuse, trafficking, smuggling, money laundering, and hawala.

Organized crime thrives in vulnerable states due to several factors:

- **Economic and Social Inequality**: High unemployment and social disparities push the poor towards organized crime.
- **Judicial Delays**: Weak witness protection and judicial delays prevent timely deterrence.
- **Political Criminalization**: Politician-criminal nexus for vote-bank politics hinders strict action against criminals.
- **Centre-State Cooperation**: Lack of cooperation between central and state governments weakens enforcement.
- **State Capacity**: Limited state capacity to enforce law and order facilitates organized crime.
- **Intimidation**: Organized groups intimidate the masses, deterring complaints and witness participation.

Organized crime contributes to conflict by:

- **Violence and Unrest**: It leads to societal violence and unrest.
- **State Legitimacy**: Challenges state legitimacy and encourages counter groups.
- **Economic Loss**: Activities like money laundering and smuggling result in state revenue loss.
- **Youth Misguidance**: Misguides youth towards quick success through illegal means.

- **Corruption and Inefficiency**: Breeds corruption, crime, and inefficiency.
- **Terrorism Support**: Provides support and resources to terrorist groups.

Mains reasons behind organized crime in India are:

- There is no central legislation specifically governing organized crime in India.
- It is coupled by lack of proper resources, technology with the law enforcement agencies
- Lack of coordination between center and the states.

Thus, pressing priority should be to implement Second ARC Recommendation: India needs to enact a central law to deal with all organized crimes comprehensively. Specific provisions to define organized crime should be included on the lines of the MCOCA 1999, GCTOC 2015 etc. for a nationwide strategy against organized crime.

Q10: India's vast coastline and inbuilt deficiencies have made coastal security daunting task. Elaborate.
(10 marks, 150 words)

Approach to the answer:
Introduction: Start with introducing in brief about India's coastline and its importance in brief.
Body: Address the demand by highlighting key deficiencies of India's coastal security followed by the measures to address those deficiencies.
Conclusion: Conclude with summarizing the topic.

Answer: India's 7,500 km coastline, crucial for national security, economic resources, and sea-borne trade, faces significant threats from hostile neighbors like China and Pakistan, as well as non-state actors such as pirates, smugglers, and terrorists.

Key deficiencies in India's coastal security include:

- **Multiple Agencies**: Lack of accountability and coordination among numerous agencies hampers seamless cooperation.
- **Absence of Apex Maritime Authority**: There is no central authority to ensure sectoral division and streamline functions.
- **Infrastructure and Resource Shortages**: Delays in infrastructure development, under-utilization of patrol boats, manpower shortages, and unspent funds.
- **Port Security**: An IB audit in 2016 revealed that out of 227 minor ports, 187 had little or no security.
- **Monitoring Deficiencies**: No effective mechanism to monitor smaller boats, many of which lack transponders.

Steps Needed:

- **Enhanced Surveillance and Coordination**: Mandatory fitment of AIS on power-driven vessels over 10 m in length and stronger inter-agency coordination.
- **Stronger Coastal Police Involvement**: Better integration and strengthening of coastal police in the littoral security framework.
- **Robust Tracking System**: All fishing vessels should be fitted with transponders.
- **Formulation of Maritime Policy**: A comprehensive policy encompassing civilian and military domains for better management of coastal waters.

The establishment of a National Maritime Domain Awareness Centre in Gurugram for coastal surveillance and monitoring is a step in the right direction. Supplementing this with proper training, funding, evaluation, monitoring, and technologies like AI and IoT will ensure a robust coastal security framework.

ADDITIONAL INFORMATION:

Importance of coastal security:

- **National Security:** Unsecured coasts threaten the mainland, as evidenced by incidents like the 26/11 attacks and Rajiv Gandhi's assassins arriving by sea.
- **Economic Resources and Activities:** Coastal security protects offshore oil and gas rigs, refineries, shipyards, and major cities such as Mumbai and Chennai.
- **Sea-borne Trade:** With 90% of India's trade occurring via coastal routes, secure coastlines are vital for economic stability.
- **Threats from Non-State Actors:** Increasing activities by pirates, smugglers, traffickers, terrorists, and illegal migrants make coastal security essential.
- **Geopolitical Concerns:** Rising Chinese assertiveness in the Indian Ocean region underscores the need for robust coastal defense.

Q11: Discuss the concept of gene therapy and its potential benefits for treating genetic diseases. Also, highlight the challenges associated with gene therapy and suggest measures to address these challenges.

(15 marks, 250 words)

Approach to the answer:
Introduction: Start with introducing in brief about Gene therapy.
Body: Address the demand by highlighting advantages of promoting gene therapy followed by Challenges associated with Gene Therapy.
Conclusion: Conclude with summarizing the topic.

Answer: Gene therapy aims to address hereditary diseases through the introduction, removal, or alteration of genetic material with the goal of treating or potentially curing these conditions. This approach seeks to provide long-term or permanent solutions for genetic disorders.

Gene Therapy Products (GTPs) involve various mechanisms for delivering nucleic acid components to achieve therapeutic benefits. These mechanisms include gene augmentation, gene editing, gene silencing, and synthetic gene enhancement. Etc.

Advantages of Promoting Gene Therapy

- **Permanent Treatment for Genetic Diseases**: Gene therapy offers the potential for long-lasting, sometimes permanent, treatment outcomes for genetic disorders.
- **High Burden of Rare Genetic Diseases in India**: With approximately 7 crore people in India suffering from rare genetic diseases, gene therapy could be transformative for these patients.
- **Market Potential and Medical Tourism**: The global gene therapy market is projected to reach up to $250 billion by 2025, presenting opportunities for medical tourism and economic growth.
- **Global Advancements**: Successful approvals of gene therapy products in the USA and Europe have paved the way for treating previously untreatable disorders. Adopting similar guidelines in India could accelerate the development and approval of gene therapies.

Challenges associated with Gene Therapy:
Technical Challenges:

- **Immune Reactions**: Gene therapy can provoke unwanted immune responses, such as when viral vectors used for delivery are attacked by the body's immune system.
- **Incorrect Targeting**: Current gene therapy techniques may sometimes target the wrong cells.
- **Virus Mutation**: Delivery viruses might mutate and potentially cause harm.

Ethical Challenges:

- **Ethical Concerns**: The creation of genetically modified embryos, as demonstrated by a Chinese scientist, has sparked global debate over the ethical implications of gene editing.

- **"Playing God" Debate**: The potential to alter human genetics raises philosophical and ethical questions about the extent of human intervention in natural processes.

Way Forward

The 2019 guidelines for gene therapy in India represent an initial step towards advancing GTPs in the country. To become a global leader in gene therapy, India needs to focus on:

- **Increased Investment**: Boost funding for gene therapy research and development.
- **R&D Support**: Enhance support for research and development to foster innovation.
- **Education and Awareness**: Improve education and raise awareness among the clinical community about gene therapy to ensure effective implementation and utilization.

By addressing these challenges and leveraging opportunities, India can advance its capabilities in gene therapy and provide new treatment options for patients with genetic diseases.

Q12: 'Biotechnology has the potential to make agricultural practices sustainable.' In this context, what are the potential benefits and risks associated with adopting genetically modified (GM) crops?

(15 Marks, 250 Words)

Approach to the answer:
Introduction: Start with introducing in brief about application of biotech in agriculture.
Body: Address the demand by highlighting examples of GM crops followed by potential benefits and challenges.
Conclusion: Conclude with summarizing the topic or with a way forward.

Answer: Biotechnology in agriculture, particularly through genetically modified (GM) crops offers innovative solutions to enhance crop resilience and productivity. By integrating specific traits, biotechnology addresses challenges like climate change, pests, and nutritional deficiencies.

Examples of GM crops:

- **GM Rubber:** GM Rubber is engineered to withstand environmental stresses like drought and cold, promoting sustainability by reducing resource dependency.
- **Bt Cotton:** Incorporates the cry1Ac gene, making it pest-resistant and reducing the need for chemical pesticides, thus lowering environmental pollution.
- **Golden Rice:** Enriched with Vitamin A, Golden Rice addresses nutritional deficiencies, promoting food security.
- **GM Mustard (DMH-11):** This hybrid crop promises higher yields and reduced reliance on imported oils, enhancing self-sufficiency.

Potential Benefits

- **Increased Yield and Productivity:** GM crops like Bt Cotton and GM Mustard can result in higher crop yields, contributing to food security and economic benefits for farmers.
- **Reduced Chemical Use:** Pest-resistant crops like Bt Cotton reduce the need for harmful pesticides, minimizing soil and water contamination.
- **Enhanced Nutritional Value:** Crops like Golden Rice can help combat malnutrition by providing essential nutrients.
- **Climate Resilience:** Crops engineered to withstand environmental stresses can help maintain agricultural productivity in the face of climate change.

Potential Risks and Challenges

- **Environmental Impact:** There are concerns of affecting biodiversity, as seen with fears of cross-pollination in Bt Brinjal and GM Mustard.
- **Health Concerns:** The long-term health impacts of consuming GM foods are not fully understood, it requires cautious evaluation.

- **Resistance Issues:** Pests develop resistance to GM crops leading to new agricultural challenges as observed with Bt Cotton and the pink bollworm.
- **Illegal Cultivation:** The unauthorized cultivation of GM crops such as the reported cases of Bt Brinjal, which poses regulatory and safety concerns.

Way Forward:

- **Comprehensive Scientific Evaluation:** Conduct rigorous and transparent assessments of GM crops, including long-term studies on health and environmental impacts.
- **Strengthened Regulatory Framework:** Establish robust mechanisms to prevent the illegal distribution and cultivation of GM crops, including stringent labeling requirements and awareness programs for farmers.
- **International Collaboration:** Learn from international experiences with GM crops to inform domestic policies and practices.
- **Public Awareness and Education:** Enhance public understanding of biotechnology in agriculture to enable informed decision-making.

While GM crops offer significant potential for sustainable agriculture, careful consideration of their risks and benefits is essential. A balanced approach, integrating scientific rigor and regulatory oversight, can ensure safe and beneficial applications of biotechnology in agriculture.

Q13: Explain the concept of Quantum Supremacy and highlight its potential applications across various sectors. Also, discuss the challenges faced in achieving practical applications of quantum computing.

(15 Marks, 250 Words)

Approach to the answer:
Introduction: Start with introducing in brief about Quantum supremacy.
Body: Address the demand by highlighting its potential application across various sectors followed by challenges associated with it.
Conclusion: Conclude with summarizing the topic or a way forward.

Answer: Quantum Supremacy refers to the point where quantum computers can perform a calculation that is beyond the reach of classical computers.

Google claimed that Sycamore was able to complete a complex computation involving the generation of a sequence of numbers in 200 seconds—a task that would take the most powerful classical supercomputers approximately 10,000 years to complete.

Potential Applications Across Various Sectors

- **Military and Security**: Quantum computers have the potential to break advanced encryption methods, enabling the decryption of sensitive communications through brute force searches. Additionally, quantum mechanics can be used to create highly secure private keys for encrypting messages.
- **Cryptography**: The inherent uncertainties in quantum mechanics can be harnessed to develop advanced cryptographic systems, ensuring secure data transmission.
- **Climate Change and Weather Forecasting**: Quantum computing could significantly improve the accuracy and speed of climate models and weather forecasts, aiding in better decision-making and disaster preparedness.
- **Data Analysis and Optimization**: In industrial science and business, quantum computers can analyze vast amounts of data more efficiently, providing quicker solutions to complex problems, especially those involving big data.
- **Healthcare and Medicine**: Quantum computing could revolutionize drug development and discovery by simulating molecular structures and interactions at an unprecedented scale and speed, leading to faster and more cost-effective medical treatments.

- **Material Science and Engineering**: Quantum computers can model complex materials and chemical reactions, aiding in the discovery of new materials with unique properties for various applications.
- **Machine Learning**: Enhanced optimization algorithms can improve the efficiency, accuracy, and speed of machine learning models, leading to advancements in artificial intelligence.
- **Teleportation of Information**: Quantum entanglement could allow for the teleportation of information, transferring data from one location to another without physical transmission.

Challenges in Achieving Practical Quantum Computing

- **Quantum Decoherence**: Quantum systems are extremely sensitive to environmental disturbances. The slightest interference, such as a stray photon or electromagnetic radiation, can cause the quantum state to collapse, a phenomenon known as quantum decoherence. This instability poses a major obstacle to reliable quantum computations.
- **Scalability**: Building quantum computers with a large number of qubits while maintaining low error rates is a significant technical challenge.
- **Resource Intensity**: Quantum computers often require extreme conditions, such as ultra-low temperatures, to function properly. This demands specialized infrastructure and significant energy consumption.
- **Standardization and Integration**: The field lacks standardized protocols and integration frameworks for quantum and classical computing systems, complicating practical deployment.
- **Manpower and Training**: The specialized knowledge required to develop and operate quantum systems necessitates a highly skilled workforce, which is currently limited.
- **Policy and Governance**: As quantum computers handle new types of data and computations, there is a need for updated legal frameworks addressing data privacy, security, and ethical considerations.

Continued investment in research, development, and training, along with international collaboration and supportive policies, will be essential to overcome these obstacles and harness the transformative power of quantum computing.

Q14: Discuss the causes and consequences of groundwater depletion in India. What measures can be taken to promote sustainable groundwater management? (15 marks, 250 words)

Approach to the answer:
Introduction: Introduce the answer by stating facts about the groundwater extraction in India.
Body: Mention the reasons and effects of groundwater depletion on all aspects of the environment. Suggest measures that can be taken to promote sustainable groundwater management.
Conclusion: Conclude by summarizing the key point stressing the importance of implementing the measures for long-term sustainability.

Answer: India is the largest user of ground water in the world, extracting 253 BCM per year, which is 25% of the global ground water extraction. This indicates severe depletion and unsustainable water management practices.

Causes of groundwater depletion in India

- Over-extraction for Agriculture as groundwater is the dominant source of irrigation.
  - Irrigation accounts for about 90% of the total groundwater extracted in India.
- Destruction of wetlands and aquifers which are used to act as water sinks and contributed to ground water recharge.
- Rapid Urbanization and Industrialization have put immense pressure on groundwater reserves.
- Climate Change impacts groundwater through impacting precipitation, leakage from surface water, sea water intrusion into coastal aquifers.
- Inadequate regulation and enforcement of groundwater extraction have exacerbated the issue.

Consequences of groundwater depletion

- Ecological Damage occurs as lower groundwater levels affect ecosystems, like wetlands, that depend on groundwater.
- Land subsidence resulting into infrastructure damage due to excessive groundwater extraction.
  - E.g. Central Ground Water Board identified excessive groundwater extraction as one of the underlying consequences of Joshi math land subsidence (2023).
- Water shortages for domestic use are due to the depletion of aquifers.
- Seawater intrusion in coastal areas affecting the quality and usability of freshwater resources.
- Effect on economy due to increased pumping costs.

Measures for sustainable groundwater management

- Controlling over-extraction by rationalizing energy pricing to make it expensive to withdraw water.
- Efficient Irrigation techniques such as micro irrigation should be promoted.
- Development of high yielding varieties of crops can encourage farmers to move away from water guzzlers.
- Watershed Management must be promoted to curb the water scarcity problem.
- Rainwater Harvesting practices should be encouraged in urban areas.

To ensure the long-term availability of groundwater in India, promoting sustainable management requires a comprehensive strategy that combines technological solutions, regulatory measures, and community-driven initiatives.

Q15: Discuss the ecological significance of mangrove forests and examine both natural and anthropogenic threats to mangrove ecosystems in India. (15 Marks, 250 Words)

Approach to the answer:
Introduction: Start with introducing in brief about Mangroves.
Body: Address the demand by highlighting its ecological significance followed by threats related to it then give way forward.
Conclusion: Conclude with summarizing the topic.

Answer: Mangroves are a diverse group of approximately 70 species of trees, shrubs, and ferns adapted to saline, oxygen-poor coastal habitats.

Ecological significance of mangrove forests

- **Protection:** They act as natural barriers against tsunamis, storm surges, and soil erosion.
- **Sediment Accretion:** Mangroves enhance sediment deposition, contributing to land formation through sediment trapping.
- **Pollution Reduction:** They help reduce seawater pollution.
- **Biodiversity:** Mangroves provide vital breeding grounds for numerous fish species and other marine fauna.
- **Economic Value:** They support coastal communities by providing resources such as honey, tannins, wax, and fisheries.
- **Climate Change Mitigation:** Mangroves serve as significant carbon sinks, helping to combat climate change.

Threats

- **Natural Factors:**
  - **Climate Change:** Increased frequency and intensity of cyclones and warmer ocean waters affect seed dispersal.
  - **Infestation:** Pests such as barnacles and wood borers can damage mangrove forests.
- **Anthropogenic Threats:**
  - **Land Reclamation:** Conversion of mangrove areas for agriculture and aquaculture.

- **Industrialization and Urban Development:** Expansion along coastlines often leads to mangrove destruction.
- **Pollution:** Discharge of untreated sewage and industrial effluents.
- **Poor Implementation of Regulations:** Inadequate enforcement of Coastal Regulation Zone (CRZ) norms.
- **Invasive Species:** Introduction of non-native species that disrupt mangrove ecosystems.

Way Forward

- **Vulnerability Mapping:** Identify and map vulnerable mangrove zones.
- **Enhanced Monitoring:** Conduct land and aerial surveys to monitor mangrove health.
- **Regulation Enforcement:** Strictly implement Coastal Regulation Zone norms.
- **Protected Areas:** Establish and manage Wildlife Sanctuaries (WLS) and National Parks (NP) to safeguard mangrove ecosystems.

Government is taking proactive steps like MISHTI, Coastal Regulation Zone (CRZ) which aims to enhance mangrove cover along coastlines and safeguard mangrove ecosystems.

Q16: Western Ghats are to be declared as an Ecologically Sensitive Area (ESA). In this context, highlight the recommendations by the Gadgil committee to conserve western Ghats. (15 Marks, 250 Words)

Approach to the answer:
Introduction: Start with introducing in brief about Western Ghats.
Body: Address the demand by highlighting the significance of western ghats followed by challenges faced by it. Then mention recommendations given by Gadgil Committee to conserve western Ghats.
Conclusion: Conclude with summarizing the topic.

Answer: Recent incidents, such as the severe landslides in Wayanad, Kerala, underscore the pressing need for effective environmental management in the Western Ghats. Western Ghats consist of a chain of mountains running parallel to India's Western Coast and passing from the states of Kerala, Maharashtra, Goa, Gujarat, Tamil Nadu and Karnataka.

Significance of Western Ghats:

- **Ecological Significance:** The Western Ghats are biodiversity hotspots, harboring a unique variety of flora and fauna, many of which are endemic to the region.
- **Watershed Functions:** The Western Ghats act as a vital watershed for several major rivers, including the Godavari, Krishna, and Kaveri. Protecting these areas helps ensure the sustainability of water resources.
- **Climate Regulation:** The dense forests of the Western Ghats act as significant carbon sinks, contributing to climate regulation by absorbing large amounts of CO_2.
- **Disaster Risk Reduction:** Healthy ecosystems in the Western Ghats provide crucial services in controlling soil erosion which leads to landslides and mitigating flood risks.
- **Impact Indian Monsoon:** The Ghats influence the Indian monsoon weather patterns. They act as a barrier to rain-laden monsoon winds that sweep in from the south-west.

Threats posed by Western Ghats:

- **Deforestation Due to Agriculture Expansion**: Agricultural expansion for cash crops like tea, coffee, and rubber in the Western Ghats has led to significant deforestation and affecting biodiversity.
- **Urbanization and Infrastructure Development**: cause habitat loss, pollution, and increased human-wildlife conflict, leading to habitat fragmentation and resource depletion for wildlife.
- **Mining Activities** Mining for minerals like iron ore and bauxite causes deforestation, soil erosion, and loss of biodiversity in the Western Ghats, with significant impacts seen in areas like Karnataka and Goa.
- **Pollution from Industrial activities:** led to pollution in the Western Ghats, degrading air and water quality, particularly harming freshwater species due to untreated industrial waste and runoff.

- **Unregulated Tourism**: resulted in waste generation and unplanned development in the Western Ghats leading to deforestation and habitat loss.
- **Climate Change Impact** Climate change, exacerbated by human activities, affects the Western Ghats through altered temperature and precipitation patterns, such as flooding in Kerala.

Recommendations by the Gadgil Committee:

- **Categorization into Ecologically Sensitive Zones:** The Committee recommended designating the entire Western Ghats as an Ecologically Sensitive Area (ESA). It proposed dividing the Western Ghats into three Ecologically Sensitive Zones (ESZs) based on their ecological significance.
  - **ESZ-I:** The most sensitive areas requiring strict protection and minimal human interference.
  - **ESZ-II:** Areas that can be used for sustainable development with stringent environmental regulations.
  - **ESZ-III:** Regions where developmental activities can be undertaken with appropriate environmental safeguards.
- **Western Ghats Ecology Authority**: The committee recommended the establishment of a Western Ghats Ecology Authority (WGEA) as a statutory body under the Ministry of Environment and Forests, with powers as outlined in Section 3 of the Environment (Protection) Act, 1986.
- **Regulation of Development Activities:** Development activities such as mining, large-scale infrastructure projects, and commercial agriculture should be regulated or restricted, particularly in the more sensitive zones (ESZ-I).
- **Community Participation:** The Committee recommended creating mechanisms for local participation in decision-making processes related to conservation and development.
- **Research and Monitoring:** Recommended the establishment of a comprehensive research and monitoring system to continuously assess the health of the ecosystem and evaluate the impact of conservation measures.
- **Incentives for Conservation:** Proposed providing incentives for conservation efforts, such as financial support and technical assistance to local communities engaged in protecting the environment.
- **Environmental Impact Assessment (EIA):** Stressed the importance of conducting thorough Environmental Impact Assessments (EIAs) before approving any major development projects in the Western Ghats.

The Gadgil Committee's recommendations aimed to balance environmental conservation with developmental needs, ensuring the protection of the Western Ghats' unique ecological heritage while allowing for sustainable development.

Q17: Examine the major challenges and shortcomings associated with the Environmental Impact Assessment (EIA) in India. Suggest measures to address these issues effectively. (15 Marks, 250 Words)

Approach to the answer:
Introduction: Start with introducing in brief about Environmental Impact Assessment (EIA).
Body: Address the demand by highlighting the background of EIA in India and its legislative framework. Then mention the challenges and shortcoming in it followed by way forward.
Conclusion: Conclude with summarizing the topic.

Answer: Environmental Impact Assessment (EIA) is a process used to evaluate the environmental, socio-economic, cultural, and health impacts of proposed projects or activities. EIA serves as a decision-making tool that helps compare various project alternatives to ensure the best balance between economic and environmental costs and benefits.

Background of EIA in India

- **Early Beginnings**: The Indian EIA process began in 1976-77 when the Planning Commission tasked the Department of Science and Technology with examining river valley projects from an environmental perspective.

- **Legislative Milestones**: Until 1994, EIA was an administrative measure without statutory backing. The 1994 EIA Notification under the Environment Protection Act (EPA), 1986, made environmental clearance mandatory for certain projects, marking a significant shift in policy.

EIA Notification, 2006

- **Overview**: The EIA Notification of 2006, issued by the Ministry of Environment, Forest and Climate Change (MoEF&CC) under the EPA, 1986, aims to assess and manage the environmental impacts of developmental projects.
- **Key Features**:
  - **Categorization**: Projects are classified into Category A (requiring national-level clearance) and Category B (requiring state-level clearance). Category B is further divided into B1 (requiring EIA) and B2 (exempt from EIA).
  - **Mandatory Clearance**: Projects in both categories must obtain prior environmental clearance before construction and operation.
  - **Public Consultation**: Category A projects must undergo public consultation at both the draft and final EIA report stages.
  - **Expert Appraisal Committee (EAC)**: EACs review EIA reports and recommend or reject clearances based on their findings. They frame the scope of EIA studies and oversee public consultation processes.

Challenges and Shortcomings of EIA in India

- **Delays**: The EIA process often leads to project delays, increasing costs.
- **Limited Coverage**: Not all projects are required to undergo EIA, leaving some with significant environmental impacts unassessed.
- **Late Public Hearing**: Public hearings are often conducted at later stages rather than at the beginning, potentially increasing conflicts.
- **Quality Issues**: EIA reports sometimes contain incomplete or inaccurate data, and consultants may lack accountability.
- **Credibility Concerns**: The project proponent often funds EIA consultants, leading to potential conflicts of interest.
- **Inadequate Compliance Monitoring**: Administrative challenges hinder effective compliance with clearance conditions.
- **Exemptions**: Recent EIA rules have increasingly exempted various projects from prior clearance.

Way Forward:

- **Expand EIA Coverage**: Include all projects with substantial environmental impacts under the EIA regime.
- **Conditional Clearances**: Implement automatic withdrawal of clearance if conditions are violated.
- **Independent EIA Authority**: Establish an independent body to enhance the credibility of the EIA process.
- **Consultant Payment Reforms**: Fund EIA consultants through MoEF&CC or an independent pool to reduce conflicts of interest.
- **Enhanced Public Hearing**: Conduct public hearings at the beginning and extend their scope to include more projects.
- **Increase Transparency**: Ensure local communities have access to all relevant project information from notification to clearance.
- **Diversify the National Green Tribunal**: Include more experts from environmental fields to strengthen the tribunal's effectiveness.

The commitment to continuous improvement in EIA practices will ensure that development is both economically beneficial and environmentally responsible, thus paving the way for a more sustainable future.

Q18: Highlight the institutional framework of disaster management in India and discuss the issues associated with it. Also, suggest measures to address these issues. (15 Marks, 250 Words)

Approach to the answer:
Introduction: Start with introducing in brief about Disaster Management Act, 2005.
Body: Address the demand by highlighting the Institutional framework of disaster management in India followed by Limitations Associated with the Disaster Management Act then conclude with way forward.
Conclusion: Conclude with summarizing the topic.

Answer: The Disaster Management Act, 2005 established a comprehensive institutional framework for disaster management at various levels of government.

Institutional framework of disaster management in India:

National Level

- **National Disaster Management Authority (NDMA)**: Chaired by the Prime Minister, the NDMA is responsible for formulating policies, and coordinating with State Disaster Management Authorities (SDMAs) to ensure a holistic approach to disaster management.
- **National Executive Committee (NEC)**: Chaired by the Home Secretary, the NEC is tasked with preparing the National Plan, monitoring the implementation of the National Policy.
- **National Institute for Disaster Management (NIDM)**: Responsible for human resource development, training, research and development, documentation, and database creation in disaster management.
- **National Disaster Response Force (NDRF)**: NDRF is responsible for responding rapidly to disaster situations and assisting civil authorities during impending disasters.

State Level

- **State Disaster Management Authority (SDMA)**: Chaired by the Chief Minister, the SDMA is responsible for formulating & implementing policies and plans for disaster management at the state level.
- **State Executive Committee**: Develops the state disaster management plan in accordance with national and state guidelines.

District Level

- **District Disaster Management Authority (DDMA)**: Chaired by the District Magistrate (DM) and includes elected representatives of local authorities as co-chairpersons. The DDMA is responsible for planning, coordinating, and implementing disaster management at the district level.

Funding Mechanism

- **Disaster Relief Fund and Disaster Mitigation Fund**: Established at district, state, and national levels to provide financial support for disaster relief and mitigation activities.

Limitations Associated with the Disaster Management Act

- **Bureaucratization and Top-Down Approach**: The Act tends to follow a top-down approach, which may overlook the contributions of local communities and civil society organizations in disaster management.
- **Poor Implementation**: The Supreme Court, in Swaraj Abhiyan vs Union of India (2016), criticized the government for the poor and slow implementation of the law.
  - For example, the National Disaster Management Fund (NDMF) has not yet been established.
- **Conflicting Institutional Structures**: There are overlapping roles between the National Emergency Committee (NEC) and the National Crisis Management Cell (NCMC), leading to confusion and inefficiencies in disaster management.
- **Lack of Clarity and Transparency**: There is insufficient clarity on how political and administrative authorities should respond to disasters, with multiple nomenclatures for severity levels, such as National Disaster, L3 Level Severity (NDM Guidelines), and Calamity of Severe Nature (DMD of Home Ministry).

- **Neglected NDMA and NEC**: Issues include vacancies in key positions, excessive executive involvement by the Ministry of Home Affairs (MoHA), and irregular meetings of the NEC, as noted in the Comptroller and Auditor General (CAG) report.
- **Funding Misappropriation**: There are concerns about the misappropriation of funds, particularly from the State Disaster Response Fund (SDRF).
- **Lack of Focus on Long-Term Recovery**: The Act primarily emphasizes preparedness, immediate relief, and infrastructure protection, neglecting the critical aspect of long-term recovery. Post-disaster relief and recovery are often managed by various ministries and departments without a unified approach.

Key Recommendations to Address Issues

- **Decentralization**: State governments should take primary responsibility for disaster management, with the central government playing a supportive role.
    - The role of local governments should be prominently integrated into disaster management efforts.
- **Ending Conflicting Institutional Structures**: The National Emergency Committee (NEC) should be abolished, as its functions overlap with those of the National Crisis Management Committee (NCMC) chaired by the Cabinet Secretary.
- **Standardizing Disaster Assessment Methodology**: The Act should define a standardized methodology for disaster categorization to clarify the responsibilities of various authorities and streamline response and relief efforts.
- **Preventing Fund Misuse**: The Act should include stringent penalties and punishments for the misuse of disaster management funds.
- **Placing Disaster Management in the Concurrent List**: Disaster management should be included in the Concurrent List to ensure effective coordination and linkages between central and state governments.
- **Integrating Long-Term Recovery with Development**: Long-term recovery should be planned alongside development initiatives, incorporating health, skill-building, and livelihood diversification schemes to ensure comprehensive recovery.

Implementing these recommendations can help address the limitations of the current framework and enhance the effectiveness of disaster preparedness and response.

Q19: The causes of Left-Wing Extremism go beyond alienation and underdevelopment. Discuss the efficacy of recent initiative taken to tackle Left-Wing Extremism. (15 marks, 250 words)

Approach to the answer:
Introduction: Start with introducing in brief about Left-Wing Extremism (LWE).
Body: Address the demand by highlighting causes of left-wing extremism followed by government initiatives to tackle Left-Wing extremism.
Conclusion: Conclude with summarizing the topic or a way forward.

Answer: The history of Left-Wing Extremism (LWE) in India dates back to the 1950s with the Naxalbari uprising. LWE involves opposition to fundamental values like the rule of law and democracy, aiming to overthrow the state through violent revolution. The movement has expanded beyond its original focus on Jal-Jungle-Jameen (land, water, and forest).

Causes of Left-Wing Extremism:

- **Ideological Factors**:
    - The ideology appeals in a society with a history of discrimination and socio-economic inequalities.
    - Many liberal intellectuals are influenced by Maoist propaganda.
- **Political Factors**:
    - Lack of true representation for marginalized communities.
    - Poor functioning of Gram Sabhas, Tribal Councils, and Autonomous Councils.

- **Social Factors**:
    - Persistent social discrimination based on caste and ethnic identities.
    - Efforts by affluent classes to maintain traditional hierarchies.
- **Economic Factors**:
    - Stark economic inequalities between urban and rural areas, and within rural areas.
    - Limited economic opportunities in hinterlands and tribal areas.
- **Administrative Factors**:
    - Lack of basic amenities like food, water, and education in tribal areas and hinterlands.
    - Poor implementation of developmental projects and corruption.

Government Initiatives to Tackle Left-Wing Extremism:

- **Modernization of Police Forces**: Enhanced equipment and training improve law enforcement capabilities against LWE activities.
- **Scheme of Fortified Police Stations**: Secure police infrastructure boosts morale and effectiveness in LWE-affected areas.
- **LWE Mobile Tower Project**: Improved communication infrastructure aids coordination and surveillance in remote areas.
- **Aspirational District Program**: Focuses on holistic development, addressing root causes of LWE by improving health, education, and infrastructure.
- **SAMADHAN Doctrine**: A multi-pronged strategy encompassing Smart Leadership, Aggressive Strategy, Motivation and Training, actionable Intelligence, Dashboard-based KPIs, Harnessing Technology, Action plan, and No access to Financing.

Efficacy of these Initiatives:

- **Reduction in Affected Districts**: According to the Ministry of Home Affairs, the number of LWE-affected districts has reduced from 72 in 2021 to 58 currently.
- **Decline in Incidents**: LWE incidents have significantly decreased. Civilian deaths have reduced by 68%.
- **Decrease in Violence**: Violent incidents have dropped by 52%, and the death of security personnel has seen a 72% decline.

Despite reduced activity in traditional areas, LWE is gaining ground in new regions like Karnataka, Kerala, and Tamil Nadu. Implementing the xaxa Committee recommendations, such as adopting more participative policies and strengthening the Forest Rights Act and Gram Sabhas, can be crucial in tackling this menace.

Q20: Highlight the role of external state and non-state actors in posing threats to India's internal security. Enumerate some of the measures taken by India to address these challenges. (15 Marks, 250 Words)

Approach to the answer:
Introduction: Start with introducing what is external state and non-state actor.
Body: Address the demand by highlighting Role of External and Non-State Actors in Creating Threats to Internal Security in India followed by steps taken by India to address these Challenges.
Conclusion: Conclude with summarizing the topic.

Answer: External state actors, particularly hostile neighbouring countries, and non-state actors, including militant and insurgent groups, play critical roles in creating and exacerbating security issues within India.

Role of External and Non-State Actors in Creating Threats to Internal Security in India
1. External State Actors:

- **Proxy War and Cross-Border Terrorism by Pakistan**: Since the fall of Bangladesh in 1971, Pakistan has supported insurgent groups in Jammu and Kashmir (J&K), including organizations like Lashkar-e-Taiba (LeT) and Jamaat-ud-Dawa (JuD).
- **Support to Khalistan Movement**: In the 1980s, Pakistan supported the Khalistan movement to create a buffer state between J&K and the rest of India.

- **Support to Indian Mujahideen**: The Inter-Services Intelligence (ISI) of Pakistan used doctored videos of the Gujarat riots to mobilize, recruit, and radicalize youth.
- **China's Involvement**: China's support to some anti-India insurgent groups and its demand for scrapping the Armed Forces (Special Powers) Act have contributed to the persistence of insurgency in the North-East.

2. Non-State Actors:

- **Militant Outfits**: Groups like Lashkar-e-Taiba (LeT) have been involved in high-profile attacks such as the 2001 Indian Parliament attack and the 2008 Mumbai attacks.
- **Left-Wing Extremism**: This includes violence and lawlessness caused by groups in states like Odisha, Chhattisgarh, and Andhra Pradesh.
- **Insurgency in the North-East**: Various insurgent groups like NDFB, ULFA, and NSCN(K) are driven by issues such as illegal immigration from Bangladesh, alienation, and inadequate development.
- **Coastal and Airspace Vulnerability**: Events such as the Mumbai attacks and the Purulia arms drop highlight the growing vulnerabilities in India's coastline and airspace.

Steps Taken by India to Address These Challenges:

- **Creation of National Security Guard (NSG)**: Established after the assassination of Indira Gandhi, the NSG is a special commando force designed to counter and neutralize terrorist activities.
- **Revised Security Architecture**: Post-26/11, India set up the National Investigation Agency (NIA) under the NIA Act of 2008 and created the Steering Committee for Review of Coastal Security (SCRCS) in 2013, along with State and District level Coastal Security Committees in 2016.
- **National Command Control Communication and Intelligence (NC3I) Network**: The NC3I network includes the Information Management and Analysis Centre (IMAC) and the Information Fusion Centre (IFC) at Gurugram, serving as the nerve center for national intelligence.
- **Expansion of NSG Hubs**: New NSG hubs have been established in Mumbai, Kolkata, Chennai, and Hyderabad.
- **Unlawful Activities (Prevention) Amendment Act, 2019**: This act enhances the effectiveness of counter-terrorism measures and addresses unlawful activities by individuals and associations.
- **Amendment to the NIA Act, 2019**: Expanded the powers and jurisdiction of the NIA to enhance its capabilities.
- **Operationalization of Crime and Criminal Tracking Network & Systems (CCTNS)**: Designed to capture and analyze crime and criminal data across police stations in India.
- **National Intelligence Grid (NATGRID)**: Developed to enhance India's counter-terrorism capabilities through advanced technology.

Addressing these challenges requires a comprehensive approach that includes enhanced security measures, legislative reforms, and improved intelligence capabilities. By continuing to strengthen these areas, India aims to safeguard its internal security and maintain stability.